6.2 Gauss's 17-gon

WORKSHEET

The question

STEP II 1991, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Why does the first part, about $\cos\frac{2\pi}{5}$, appear in a question about the 17-gon?

STAGE 1 The pentagon

Ask yourself Expand $y^2 = (\cos\phi + \cos 2\phi)^2$, turn each product into a sum, and use $5\phi = 2\pi$ to reduce angles like 3ϕ and 4ϕ.

STAGE 3 Splitting the cosines into two groups

Ask yourself Let $z = \cos\theta + \cos 2\theta + \cos 4\theta + \cos 8\theta$, and let z' be the sum of the other four cosines, $\cos 3\theta + \cos 5\theta + \cos 6\theta + \cos 7\theta$. What is $z + z'$?

Pause and try Compute zz'. Expand it as 16 products $\cos a\theta \cos b\theta$, turn each into a sum of two cosines, and count how often each of $\cos\theta, \cos 2\theta, \ldots, \cos 8\theta$ appears. (Reduce angles using $\cos k\theta = \cos(17-k)\theta$ and $\cos(-k\theta) = \cos k\theta$.)

Try this next

1. **The heptagon.** Try STEP II 1995, Question 6 (step.maths.org/questions/95-s2-q6). It applies Gauss's idea of grouping roots of unity to the seventh roots, and finds exact values for sums of cosines and sines. Deciding the sign of the sine sum is the subtle step.

2. **The pentagon.** Show that $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$, and use it to describe a ruler-and-compasses construction of a regular pentagon.

3. **The other period.** Check numerically that $z' = \cos 3\theta + \cos 5\theta + \cos 6\theta + \cos 7\theta = \frac{-1-\sqrt{17}}{4}$.

4. **Generators.** Show that the powers of 3 modulo 17 give every non-zero residue, but the powers of 2 do not. Which residues do the powers of 2 give?

Further reading

- **Ian Stewart, *Galois Theory*** (CRC Press). Chapter on ruler-and-compass constructions and the regular 17-gon.

- **Richard Courant & Herbert Robbins, *What Is Mathematics?*** (Oxford University Press). Chapter 3 is a beautiful elementary account of constructions, including the 17-gon.
- **Daniel Kehlmann, *Measuring the World*** (Quercus). A comic novel about Gauss and Alexander von Humboldt, for light relief.

The full guided solution, and what the question leads to at university, is in Section 6.2:
https://beyondstep.keithmcnulty.org/ch06/gauss-17gon.html